

where the x_i values are the solutions of the equation $y = g(x)$.

2.4 MULTIPLE RANDOM VARIABLES

We have already defined the probability distribution function of an RV. For example, if X and Y are RVs, then their distribution functions are defined as

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ F_Y(y) &= P(Y \leq y) \end{aligned} \quad (2.47)$$

Now we define the probability that both $X \leq x$ and $Y \leq y$ as the joint probability distribution function of X and Y :

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) \quad (2.48)$$

If the meaning is clear from the context, we often use the shorthand notation $F(x, y)$ to represent the distribution function $F_{XY}(x, y)$. Some properties of the joint distribution function are

$$\begin{aligned} F(x, y) &\in [0, 1] \\ F(x, -\infty) = F(-\infty, y) &= 0 \\ F(\infty, \infty) &= 1 \\ F(a, c) &\leq F(b, d) \quad \text{if } a \leq b \text{ and } c \leq d \\ P(a < x \leq b, c < y \leq d) &= F(b, d) + F(a, c) - F(a, d) - F(b, c) \\ F(x, \infty) &= F(x) \\ F(\infty, y) &= F(y) \end{aligned} \quad (2.49)$$

Note from the last two properties that the distribution function of one RV can be obtained from the joint distribution function. When the distribution function for a single RV is obtained this way it is called the marginal distribution function.

The joint probability density function is defined as the following derivative of the joint PDF:

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y} \quad (2.50)$$

As before, we often use the shorthand notation $f(x, y)$ to represent the density function $f_{XY}(x, y)$. Some properties of the joint pdf that can be obtained from this definition are

$$\begin{aligned} F(x, y) &= \int_{-\infty}^x \int_{-\infty}^y f(z_1, z_2) dz_1 dz_2 \\ f(x, y) &\geq 0 \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= 1 \\ P(a < x \leq b, c < y \leq d) &= \int_c^d \int_a^b f(x, y) dx dy \\ f(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\ f(y) &= \int_{-\infty}^{\infty} f(x, y) dx \end{aligned} \quad (2.51)$$

Note from the last two properties that the density function of one RV can be obtained from the joint density function. When the density function for a single RV is obtained this way it is called the marginal density function. Computing the expected value of a function $g(\cdot, \cdot)$ of two RVs is similar to computing the expected value of a function of a single RV:

$$E[g(x, y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy \quad (2.52)$$

2.4.1 Statistical independence

Recall from Section 2.1 that two events are independent if the occurrence of one event has no effect on the probability of the occurrence of the other event. We extend this to say that RVs X and Y are independent if they satisfy the following relation:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) \quad \text{for all } x, y \quad (2.53)$$

From our definition of joint distribution and density functions, we see that this implies

$$\begin{aligned} F_{XY}(x, y) &= F_X(x)F_Y(y) \\ f_{XY}(x, y) &= f_X(x)f_Y(y) \end{aligned} \quad (2.54)$$

The central limit theorem says that the sum of independent RVs tends toward a Gaussian RV, regardless of the pdf of the individual RVs that contribute to the sum. This is why so many RVs in nature seem to have a Gaussian distribution. Many RVs in nature are actually the sum of many individual and independent RVs. For example, the high temperature on any given day in any given location tends to follow a Gaussian distribution. This is because the high temperature is affected by clouds, precipitation, wind, air pressure, humidity, and other factors. Each of these factors is in turn determined by other random factors. The combination of many independent random variables determines the high temperature, which has a Gaussian pdf.

We define the covariance of two scalar RVs X and Y as

$$\begin{aligned} C_{XY} &= E[(X - \bar{X})(Y - \bar{Y})] \\ &= E(XY) - \bar{X}\bar{Y} \end{aligned} \quad (2.55)$$

We define the correlation coefficient of two scalar RVs X and Y as

$$\rho = \frac{C_{XY}}{\sigma_x \sigma_y} \quad (2.56)$$

The correlation coefficient is a normalized measurement of the independence between two RVs X and Y . If X and Y are independent, then $\rho = 0$ (although the converse is not necessarily true). If Y is a linear function of X then $\rho = \pm 1$ (see Problem 2.9).

We define the correlation of two scalar RVs X and Y as

$$R_{XY} = E(XY) \quad (2.57)$$

Two RVs are said to be uncorrelated if $R_{XY} = E(X)E(Y)$.

From the definition of independence, we see that if two RVs are independent then they are also uncorrelated. Independence implies uncorrelatedness, but uncorrelatedness does not necessarily imply independence. However, in the special case in which two RVs are both Gaussian and uncorrelated, then it follows that they are also independent.

Two RVs are said to be orthogonal if $R_{XY} = 0$. If two RVs are uncorrelated, then they are orthogonal only if at least one of them is zero-mean. If two RVs are orthogonal, then they may or may not be uncorrelated.

■ EXAMPLE 2.6

Two rolls of the dice are represented by the RVs X and Y . The two RVs are independent because one roll of the die does not have any effect on a second roll of the die. Each roll of the die has an equally likely probability ($1/6$) of being a 1, 2, 3, 4, 5, or 6. Therefore,

$$\begin{aligned} E(X) = E(Y) &= \frac{1 + 2 + 3 + 4 + 5 + 6}{6} \\ &= 3.5 \end{aligned} \quad (2.58)$$

There are 36 possible combinations of the two rolls of the die. We could get the combination (1,1), (1,2), and so on. Each of these 36 combinations have an equally likely probability ($1/36$). Therefore, the correlation between X and Y is

$$\begin{aligned} R_{XY} = E(XY) &= \frac{1}{36} \sum_{i=1}^6 \sum_{j=1}^6 ij \\ &= 12.25 \\ &= E(X)E(Y) \end{aligned} \quad (2.59)$$

Since $E(XY) = E(X)E(Y)$, we see that X and Y are uncorrelated. However, $R_{XY} \neq 0$, so X and Y are not orthogonal.

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■ EXAMPLE 2.7

A slot machine is rigged so you get 1 or -1 with equal probability the first spin X , and the opposite number the second spin Y . We have equal probabilities of obtaining (X, Y) outcomes of (1, -1) and (-1 , 1). The two RVs are dependent because the realization of Y depends on the realization of X . We also see that

$$\begin{aligned} E(X) &= 0 \\ E(Y) &= 0 \\ E(XY) &= \frac{(1)(-1) + (-1)(1)}{2} \\ &= -1 \end{aligned} \quad (2.60)$$

We see that X and Y are correlated because $E(XY) \neq E(X)E(Y)$. We also see that X and Y are not orthogonal because $E(XY) \neq 0$.

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■ EXAMPLE 2.8

A slot machine is rigged so you get -1 , 0 , or $+1$ with equal probability the first spin X . On the second spin Y you get 1 if $X = 0$, and 0 if $X \neq 0$. The two RVs are dependent because the realization of Y depends on the realization of X . We also see that

$$\begin{aligned}
 E(X) &= \frac{-1 + 0 + 1}{3} \\
 &= 0 \\
 E(Y) &= \frac{0 + 1 + 0}{3} \\
 &= 1/3 \\
 E(XY) &= \frac{(-1)(0) + (0)(1) + (1)(0)}{3} \\
 &= 0
 \end{aligned} \tag{2.61}$$

We see that X and Y are uncorrelated because $E(XY) = E(X)E(Y)$. We also see that X and Y are orthogonal because $E(XY) = 0$. This example illustrates the fact that uncorrelatedness does not necessarily imply independence.

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■ EXAMPLE 2.9

Suppose that x and y are independent RVs, and the RV z is computed as $z = g(x) + h(y)$. In this example, we will calculate the mean of z :

$$\begin{aligned}
 E(z) &= E[g(x) + h(y)] \\
 &= \int \int [g(x) + h(y)] f(x, y) dx dy \\
 &= \int \int g(x) f(x) f(y) dx dy + \int \int h(y) f(x) f(y) dx dy \\
 &= \int g(x) f(x) dx \int f(y) dy + \int h(y) f(y) dy \int f(x) dx \\
 &= E[g(x)](1) + E[h(y)](1) \\
 &= E[g(x)] + E[h(y)]
 \end{aligned} \tag{2.62}$$

As a special case of this example, we see that the mean of the sum of two independent RVs is equal to the sum of their means. That is,

$$E(x + y) = E(x) + E(y) \quad \text{if } x \text{ and } y \text{ are independent} \tag{2.63}$$

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■ EXAMPLE 2.10

Suppose we roll a die twice. What is the expected value of the sum of the two outcomes? We use X and Y to refer to the two rolls of the die, and we use

Z to refer to the sum of the two outcomes. Therefore, $Z = X + Y$. Since X and Y are independent, we have

$$\begin{aligned} E(Z) &= E(X) + E(Y) \\ &= 3.5 + 3.5 \\ &= 7 \end{aligned} \quad (2.64)$$

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■ EXAMPLE 2.11

Consider the circuit of Figure 2.4. The input voltage V is uniformly distributed on $[-1, 1]$. Voltage V has units of volts, and the two currents have units of amps.

$$\begin{aligned} I_1 &= \begin{cases} 0 & \text{if } V > 0 \\ V & \text{if } V \leq 0 \end{cases} \\ I_2 &= \begin{cases} V & \text{if } V \geq 0 \\ 0 & \text{if } V < 0 \end{cases} \end{aligned} \quad (2.65)$$

We see that I_1 is uniformly distributed on $[-1, 0]$ and I_2 is uniformly distributed on $[0, 1]$. The RVs V , I_1 , and I_2 have expected values

$$\begin{aligned} E(V) &= 0 \\ E(I_1) &= -1/2 \\ E(I_2) &= 1/2 \end{aligned} \quad (2.66)$$

The RVs I_1 and I_2 are not independent because they are related to each other; if $I_2 \neq 0$ then $I_1 = 0$, and if $I_1 \neq 0$ then $I_2 = 0$. Since either I_1 or I_2 is equal to 0 at every time instant, $I_1 I_2 = 0$ and $E(I_1 I_2) = 0$. Therefore I_1 and I_2 are orthogonal. Since $E(I_1)E(I_2) = -1/4$, we see that $E(I_1 I_2) \neq E(I_1)E(I_2)$, and I_1 and I_2 are correlated.

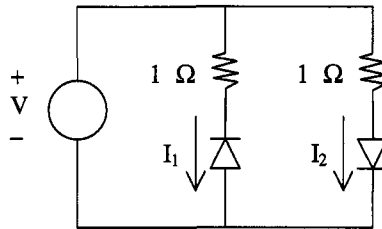


Figure 2.4 Circuit for Example 2.11.

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2.4.2 Multivariate statistics

The discussion in the previous subsection can be generalized for RVs that are vectors. In this case, the quantities defined earlier become vectors and matrices. Given

an n -element RV X and an m -element RV Y (assuming that both X and Y are column vectors), their correlation is defined as

$$\begin{aligned} R_{XY} &= E(XY^T) \\ &= \begin{bmatrix} E(X_1Y_1) & \cdots & E(X_1Y_m) \\ \vdots & & \vdots \\ E(X_nY_1) & \cdots & E(X_nY_m) \end{bmatrix} \end{aligned} \quad (2.67)$$

Their covariance is defined as

$$\begin{aligned} C_{XY} &= E[(X - \bar{X})(Y - \bar{Y})^T] \\ &= E(XY^T) - \bar{X}\bar{Y}^T \end{aligned} \quad (2.68)$$

The autocorrelation of the n -element RV X is defined as

$$\begin{aligned} R_X &= E[XX^T] \\ &= \begin{bmatrix} E[X_1^2] & \cdots & E[X_1X_n] \\ \vdots & & \vdots \\ E[X_nX_1] & \cdots & E[X_n^2] \end{bmatrix} \end{aligned} \quad (2.69)$$

Note that $E(X_iX_j) = E(X_jX_i)$ so $R_X = R_X^T$. An autocorrelation matrix is always symmetric. Also note that for any n -element column vector z we have

$$\begin{aligned} z^T R_X z &= z^T E[XX^T] z \\ &= E[z^T XX^T z] \\ &= E[(z^T X)^2] \\ &\geq 0 \end{aligned} \quad (2.70)$$

So an autocorrelation matrix is always positive semidefinite.

The autocovariance of the n -element RV X is defined as

$$\begin{aligned} C_X &= E[(X - \bar{X})(X - \bar{X})^T] \\ &= \begin{bmatrix} E[(X_1 - \bar{X}_1)^2] & \cdots & E[(X_1 - \bar{X}_1)(X_n - \bar{X}_n)] \\ \vdots & & \vdots \\ E[(X_n - \bar{X}_n)(X_1 - \bar{X}_1)] & \cdots & E[(X_n - \bar{X}_n)^2] \end{bmatrix} \\ &= \begin{bmatrix} \sigma_1^2 & \cdots & \sigma_{1n} \\ \vdots & & \vdots \\ \sigma_{n1} & \cdots & \sigma_n^2 \end{bmatrix} \end{aligned} \quad (2.71)$$

Note that $\sigma_{ij} = \sigma_{ji}$, so $C_X = C_X^T$. An autocovariance matrix is always symmetric. Also note that for any n -element column vector z we have

$$\begin{aligned} z^T C_X z &= z^T E[(X - \bar{X})(X - \bar{X})^T] z \\ &= E[z^T (X - \bar{X})(X - \bar{X})^T z] \\ &= E[(z^T (X - \bar{X}))^2] \\ &\geq 0 \end{aligned} \quad (2.72)$$

So an autocovariance matrix is always positive semidefinite.

An n -element RV X is Gaussian (normal)³ if

$$\text{pdf}(X) = \frac{1}{(2\pi)^{n/2}|C_X|^{1/2}} \exp \left[\frac{-1}{2}(X - \bar{X})^T C_X^{-1}(X - \bar{X}) \right] \quad (2.73)$$

Now consider a Gaussian RV X that undergoes a linear transformation:

$$\begin{aligned} Y &= g(X) \\ &= AX + b \end{aligned} \quad (2.74)$$

where A is a constant $n \times n$ matrix, and b is a constant n -element vector. If A is invertible, then

$$\begin{aligned} X &= h(Y) \\ &= A^{-1}Y - A^{-1}b \end{aligned} \quad (2.75)$$

From Equation (2.42) we obtain

$$\begin{aligned} f_Y(y) &= |h'(y)|f_X[h(y)] \\ &= |A^{-1}| \frac{1}{(2\pi)^{n/2}|C_X|^{1/2}} \exp \left[\frac{-1}{2}(A^{-1}y - A^{-1}b - \bar{x})^T C_X^{-1}(\dots) \right] \\ &= |A^{-1}| \frac{1}{(2\pi)^{n/2}|C_X|^{1/2}} \times \\ &\quad \exp \left[\frac{-1}{2}(A^{-1}y - A^{-1}b - A^{-1}A\bar{x})^T C_X^{-1}(\dots) \right] \\ &= \frac{1}{(2\pi)^{n/2}|A||C_X|^{1/2}} \times \\ &\quad \exp \left[\frac{-1}{2}(A^{-1}y - A^{-1}\bar{y})^T C_X^{-1}(A^{-1}y - A^{-1}\bar{y}) \right] \\ &= \frac{1}{(2\pi)^{n/2}|A|^{1/2}|C_X|^{1/2}|A^T|^{1/2}} \times \\ &\quad \exp \left[\frac{-1}{2}(y - \bar{y})^T A^{-T} C_X^{-1} A^{-1}(y - \bar{y}) \right] \\ &= \frac{1}{(2\pi)^{n/2}|AC_X A^T|^{1/2}} \exp \left[\frac{-1}{2}(y - \bar{y})^T (AC_X A^T)^{-1}(y - \bar{y}) \right] \\ y &\sim N(A\bar{x} + b, AC_X A^T) \end{aligned} \quad (2.76)$$

This shows that normality is preserved in linear transformations of random vectors (just as it is preserved in linear transformations of random scalars, as seen in Example 2.4).

³Francis Edgeworth (1845-1926), an Irish economist and mathematician, first provided a general description and study of the multivariate Gaussian probability distribution in 1892 [Sor80].